Drake Ferronyalka

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PROFESSIONAL SUMMARY

Mechanical Engineering student and U.S. Navy veteran who takes structural and mechanical hardware from concept through hand calculation, fabrication, and testing. Proficient in SolidWorks (CSWA Certified), GD&T, and DFM/DFA, with hands-on prototype machining and assembly experience. Focused on space systems for lunar surface operations.

EDUCATION

University of Central Florida **Jan 2026 – Present**

B.S. Mechanical Engineering | (Expected Graduation: May 2028; GPA: 3.85)

Activities: Knights Satellite Club; Students for the Exploration and Development of Space (SEDS)

State College of Florida **Dec 2025**

A.A. | President's List (Fall 2024, Spring/Fall 2025); Dean's List (Spring/Summer 2024)

PROJECTS

DTA-1 Dust-Tolerant Actuator **Jul 2026 – Present**

SolidWorks, Gear Train Design, BLDC Motor Selection

- Designing a 4-inch OD rotary actuator mechanism with a two-stage planetary gearbox (projected 9:1) for planetary surface environments, driven by a frameless CubeMars RI80-PH BLDC motor in a sealed housing. Evaluating enclosure temperatures, sealing architecture, and dust ingress mitigation.

X-Axis Linear Rail Conversion – CNC Plasma Table **Jan 2026 – May 2026**

SolidWorks, Hand Calculations, Linear Rail Systems, Machining

- Replaced the hand-preloaded ball bearing X-axis carriage on a Langmuir CrossFire XR plasma table with an HGR20 profile rail; validated by an FDM test article run in production for two months before machining in billet, now in continuous service with no preload corrections.
- Sized a two-part 6061-T6 bracket by hand calculation from measured loads (25.31 N Z-axis assembly, 22.27 N cableway), reacting a 1.974 N·m root moment as a force couple across two HGH20CA blocks at 78.7 mm centers, the minimum achievable spacing for the block size.

Wheel Light Mounting System (US 12,673,603 B2 – granted Jul 2026) **2024 – 2025**

SolidWorks, GD&T, Prototype Fabrication

- As a named co-inventor, designed a modular mounting architecture for off-road lighting with emphasis on vibration resistance, packaging, and serviceability; produced manufacturing-ready drawings and BOM, fabricated laser-cut / formed prototypes and assembled in-house; validated performance via torque checks and road testing.

EXPERIENCE

Machinist Apprentice **May 2026 – Present**

University of Central Florida, Office of Research *Orlando, FL*

- Manufacture research hardware on 3-axis CNC equipment from engineering drawings for faculty-led labs, working to specified tolerances. Inspect and verify finished parts against drawings using calipers and micrometers.
- Write G-code by hand for wire EDM operations, then set up and run the machine, translating part requirements into tool paths.

Mechanical Design and Fabrication Technician **2020 – Present**

AFNY Specialties (2020 – 2025) / Pulsar Designs LLC (2025 – Present) *Ellenton, FL*

- Designed and fabricated SolidWorks assemblies ranging from single-piece brackets to 10–15-part assemblies and suspension hardware, producing BOMs and assembly notes to support fabrication and installation.
- Standardized repeatable threaded features using a FlexArm tapping system, optimizing tap geometry, torque control, and lubrication practices to reduce tool breakage and maintain thread quality.

Fire Controlman / 400 Hz Power System Maintainer **2017 – 2023**

United States Navy *Various Locations*

- Maintained and troubleshot the Aegis MK 99 T-1348 Radar Transmitter and 400 Hz Air-Cooled Solid-State Frequency Converter power systems, isolating faults to component level and verifying restored performance against specified requirements.
- Led and trained 10+ technicians in diagnostics, troubleshooting workflows, and procedure compliance, standardizing checkouts to improve team execution.

SKILLS

CAD & Design: SolidWorks (CSWA Certified), Siemens NX (modeling), GD&T (ASME Y14.5), Technical Drawing, Tolerance Stack-Up, DFM/DFA

Analysis: ANSYS Static Structural, First-Principles Sizing, Shaft & Gear Load Analysis

Manufacturing: CNC 3-Axis Mill (setup & operation), Wire EDM (setup, G-code & operation), Manual Lathe & Mill, CNC Plasma (SheetCAM), Sheet Metal/Plate Fabrication, Press Brake Forming, Tapping/Threaded Features, Fixture Design, FDM, Assembly & Integration

Instrumentation & Electronics: BLDC Motor Control (VESC), Data Acquisition, Load Cells, Thermocouples, IMU, Oscilloscope, Soldering, Radar & 400 Hz Power Systems

Software: MATLAB, Python, C, Embedded (RP2040 / Raspberry Pi), Microsoft Excel/Word/PowerPoint